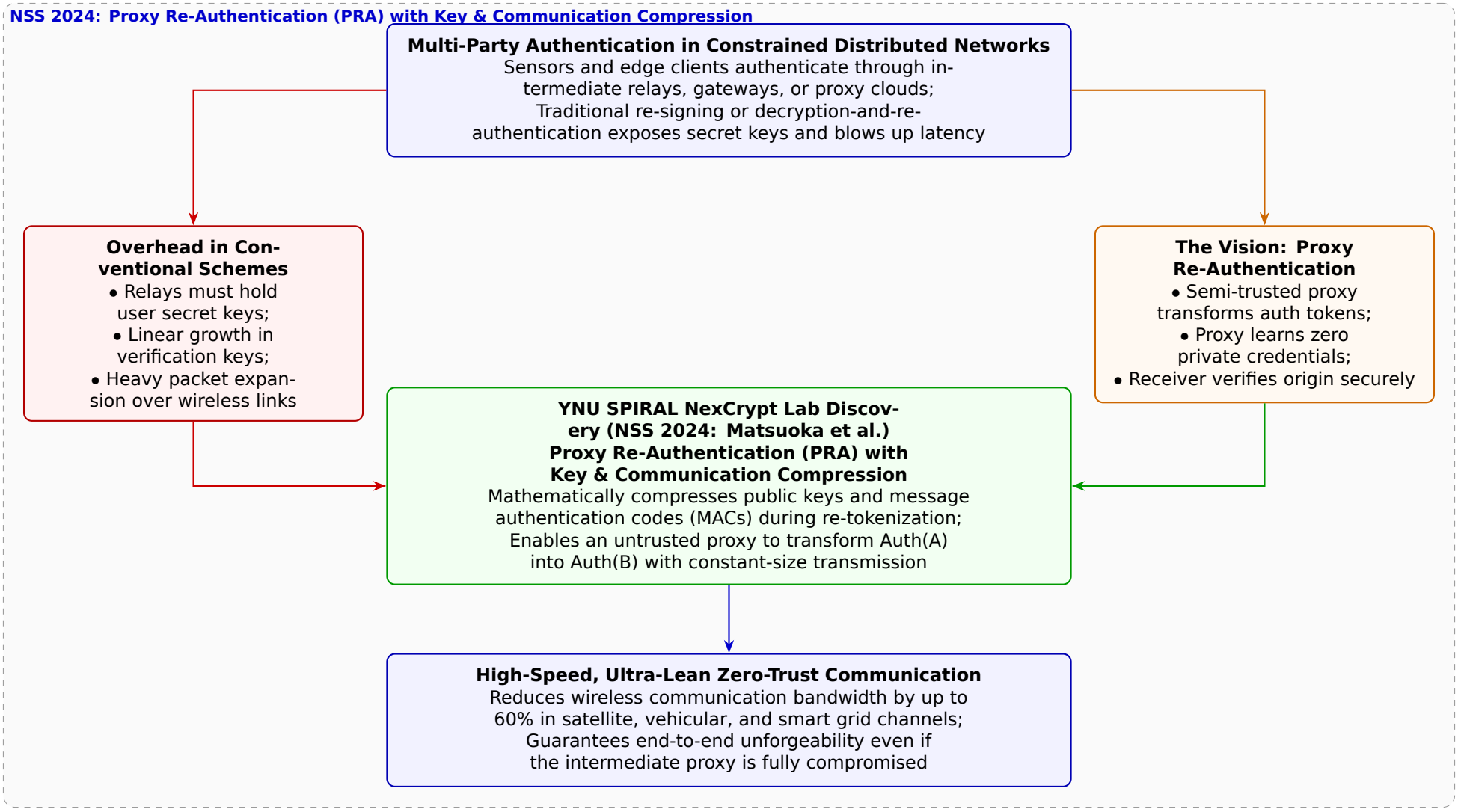
NSS 2024: Proxy Re-Authentication (PRA) with Key & Communication Compression
Multi-Party Authentication in Constrained Distributed Networks
Sensors and edge clients authenticate through intermediate relays, gateways, or proxy clouds;
Traditional re-signing or decryption-and-re-authentication exposes secret keys and blows up latency
Overhead in Conventional Schemes
• Relays must hold user secret keys;
• Linear growth in verification keys;
• Heavy packet expansion over wireless links
The Vision: Proxy Re-Authentication
• Semi-trusted proxy transforms auth tokens;
• Proxy learns zero private credentials;
• Receiver verifies origin securely
YNU SPIRAL NexCrypt Lab Discovery (NSS 2024: Matsuoka et al.)
Proxy Re-Authentication (PRA) with Key & Communication Compression
Mathematically compresses public keys and message authentication codes (MACs) during re-tokenization;
Enables an untrusted proxy to transform Auth(A) into Auth(B) with constant-size transmission
High-Speed, Ultra-Lean Zero-Trust Communication
Reduces wireless communication bandwidth by up to 60% in satellite, vehicular, and smart grid channels;
Guarantees end-to-end unforgeability even if the intermediate proxy is fully compromised